# Santiago Montaña

**Data Scientist | Data & AI Engineer | Conversational AI & Cloud Operations**

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## PROFESSIONAL SUMMARY

Data Scientist and Data & AI Engineer with a background in Electronic Engineering and a Specialization in Strategic Data Analytics (UPTC). Experienced in end-to-end data pipeline engineering (Snowflake, Apache Airflow, dbt, Meltano), predictive modeling and classification algorithms (Random Forest, XGBoost, Monte Carlo quantitative simulations, MLflow), conversational AI solutions (LLMs, autonomous agents, Model Context Protocol - MCP, ElevenLabs voice AI), and cloud infrastructure provisioning on AWS using Terraform (IaC), Docker, and FastAPI under CI/CD.

## PROFESSIONAL EXPERIENCE

### Junior Data Scientist & AI Developer

Jan 2026 – Present

*CIM DATA IA — Consultoría de Datos e Inteligencia Artificial — Boyacá / Remoto, Colombia*

- Developed predictive models and classification algorithms in Python (scikit-learn, Random Forest, XGBoost) for operational forecasting and prioritization, tracking experiments with MLflow.
- Built conversational AI solutions and autonomous agents integrating Model Context Protocol (MCP) to automate enterprise workflows and infrastructure deployment.
- Collaborated on cloud infrastructure provisioning and deployment on AWS (EC2, S3, IAM) using Terraform (IaC), ensuring reproducible and scalable environments.
- Executed quantitative Monte Carlo simulations and stress tests to validate data design effectiveness, orchestrating ETL/ELT pipelines with Python, SQL, Snowflake, dbt, and Apache Airflow under CI/CD.

### Development Assistant (Data & AI)

May 2025 – Dec 2025

*CIM DATA IA — Boyacá, Colombia*

- Analytically optimized and tuned banking chatbots and conversational AI flows, measurably improving intent classification accuracy and client response quality.
- Collaborated on building an AI voice assistant for automated loan verification in the financial domain using Python, ElevenLabs, and third-party REST APIs.
- Implemented operational process automations and standardized experimentation environments using Anaconda and virtualenv to guarantee cross-team portability.

### Data & Industrial Operations Automation Intern

May 2024 – Nov 2024

*HOLCIM S.A. — Operaciones Industriales y Planta — Nobsa, Boyacá, Colombia*

- Automated collection, cleaning, and processing of industrial plant operational data using Python scripts and Google Apps Script.
- Designed and implemented a centralized database to improve industrial asset traceability, reliability, and organization.
- Built interactive executive dashboards in Looker Studio to monitor plant productivity, equipment availability, and operational KPIs.

## TECHNICAL SKILLS

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|---|---|
| **Data Engineering & Orchestration:** | Snowflake (Cloud Data Warehouse), Apache Airflow (DAG Orchestration), dbt (SQL Transformations), Meltano (ELT Extraction), Pipelines ETL / ELT, Modelamiento de Datos & Calidad |
| **Machine Learning & Data Science:** | Scikit-learn (Random Forest, XGBoost, Ensembles), Simulaciones Cuantitativas (Monte Carlo), MLflow (Experiment Tracking & MLOps), TensorFlow & PyTorch, Análisis Estadístico & Validación de Hipótesis, Modelado Predictivo & Clasificación |
| **Conversational AI & MCP Agents:** | Model Context Protocol (MCP) Integration, Agentes Autónomos de IA, LLMs (Claude, OpenAI, Prompt Engineering), ElevenLabs (Asistentes de Voz con IA), Optimización de Chatbots Bancarios & NLP |
| **Cloud, DevOps & IaC:** | Terraform (Infrastructure as Code - IaC), AWS (EC2, S3, IAM), FastAPI & Pydantic (APIs REST en Producción), Docker & Entornos Reproducibles (Anaconda, virtualenv), CI/CD con GitHub Actions & Git |
| **Programming Languages:** | Python (pandas, NumPy, SciPy, scikit-learn, Matplotlib), SQL (Consultas Analíticas, CTEs, Optimización), R (Estadística & Visualización), JavaScript / TypeScript (ES6+), C++ |
| **Databases & Warehousing:** | Snowflake, PostgreSQL, MySQL, SQLite |

**Visualization & BI:** Looker Studio, Power BI, Tableau, Storytelling con Datos & KPIs Operativos

## FEATURED PROJECTS

**Autonomous News & Media AI Agent** *Stack: Python, LLMs, MCP, Autonomous Agents, Automation*

Autonomous AI pipeline integrating LLMs and structured workflows for automated news research, synthesis, and multi-channel content generation.

**Multiclass Vehicle Classification API & Web App** *Stack: Python, scikit-learn, FastAPI, Flask, Streamlit, Pydantic*

End-to-end vehicle classification model serialized and deployed with FastAPI/Flask and Streamlit, including Pydantic validation, logging, and automated testing.

**USD/COP Exchange Rate (TRM) & Crypto Monte Carlo Predictor** *Stack: Python, Scikit-learn, Monte Carlo, NLP Sentiment, Jupyter Notebook*

Predictive modeling combining historical time series, real-time news sentiment analysis, and Monte Carlo quantitative simulations to estimate market scenarios.

**Conversational Voice AI Agent (ElevenLabs + LLMs)** *Stack: Python, ElevenLabs, LLMs, Voice AI, REST APIs*

Interactive voice AI agent built with Python and ElevenLabs APIs for low-latency automated verification and natural conversational workflows.

## EDUCATION

**Specialization in Strategic Data Analytics**
*Universidad Pedagógica y Tecnológica de Colombia (UPTC)*

**Bachelor of Science in Electronic Engineering**
*Universidad Pedagógica y Tecnológica de Colombia (UPTC)*

## LANGUAGES

**Spanish** — Native (C2)

**English** — Upper-Intermediate (B2 professional reading & technical documentation)

## CERTIFICATIONS & COURSES

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| MLOps & Despliegue de Modelos de Machine Learning | 2025 | Data Scientist con Python | Apr 2025 |
| Decision Trees y Random Forest con scikit-learn | Mar 2025 | Tableau — Visualización de Datos y Storytelling | Mar 2025 |
| Regresión Lineal con Python y scikit-learn | Feb 2025 | Crear tus Proyectos de Ciencia de Datos | Feb 2025 |
| Introducción a Machine Learning | Jan 2025 | Cálculo Diferencial para Data Science | Jan 2025 |
| Git y GitHub Profesional | Jan 2025 | Estadística Descriptiva para Data Science | Dec 2024 |
| Jupyter Notebooks y Anaconda para Ciencia de Datos | Nov 2024 | Bases de Datos con SQL | Sep 2024 |
| Python para Ciencia de Datos | Sep 2024 | Power BI para Analítica de Negocios | 2025 |